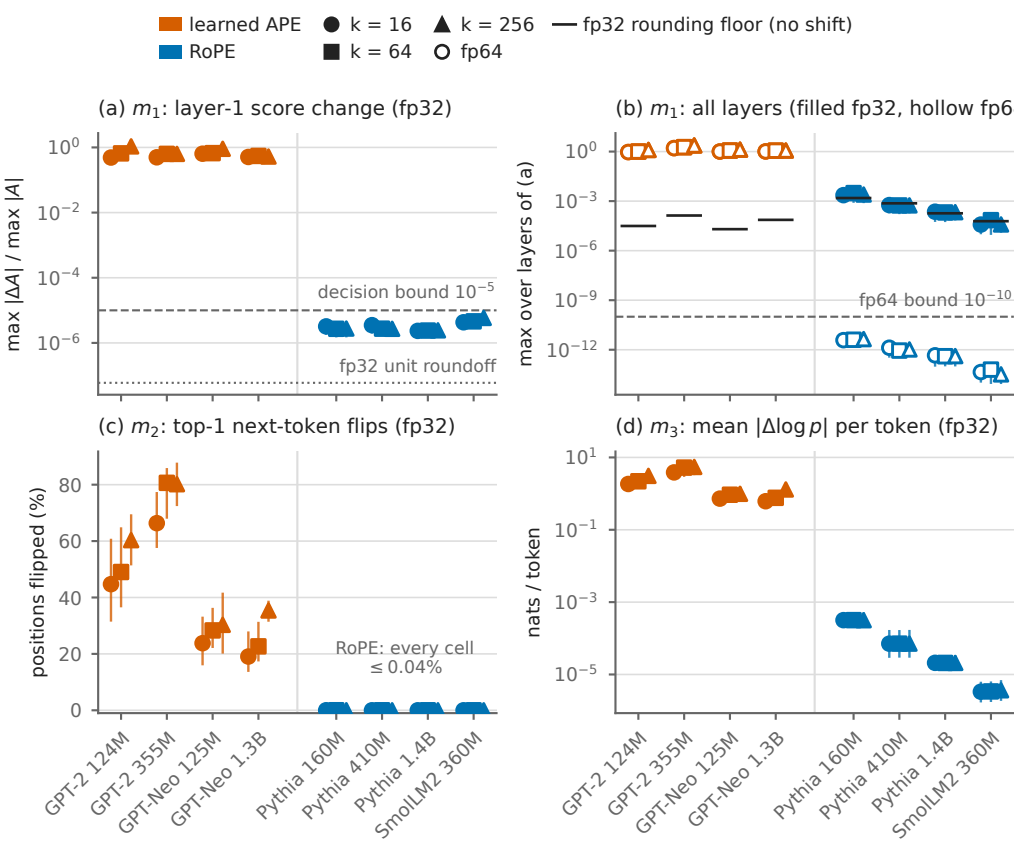
learned APE
RoPE
k = 16
k = 64
k = 256
fp64
fp32 rounding floor (no shift)
(a) m_1: layer-1 score change (fp32)
max |ΔA| / max |A|
decision bound 10^{-5}
fp32 unit roundoff
(b) m_1: all layers (filled fp32, hollow fp64)
max over layers of (a)
fp64 bound 10^{-10}
(c) m_2: top-1 next-token flips (fp32)
positions flipped (%)
RoPE: every cell ≤ 0.04%
(d) m_3: mean |Δlog p| per token (fp32)
nats / token
GPT-2 124M
GPT-2 355M
GPT-Neo 125M
GPT-Neo 1.3B
Pythia 160M
Pythia 410M
Pythia 1.4B
SmolLM2 360M